Finding Limits Analytically

These types of examples will come up when I cover Section 2.6.

Example 1: Find the following limit **analytically**. If the limit is ∞ or −∞, then state that.

$$\lim_{x\to-\infty} \tan[\csc^{-1}(e^{-x})] = 0$$

$$x \to -\infty \Rightarrow e^{-x} \to \infty \Rightarrow \csc^{-1}(e^{-x}) \to 0 \Rightarrow \tan[\csc^{-1}(e^{-x})] \to \tan(0) = 0$$

Example 2: Find the following limit **analytically**. If the limit is ∞ or −∞, then state that.

$$\lim_{x\to\infty} \sin[\tan^{-1}(-\ln(x^2)]$$

$$x \to \infty \Rightarrow x^2 \to \infty \Rightarrow -\ln(x^2) \to -\infty \Rightarrow \tan^{-1}(-\ln(x^2)) \to -\frac{\pi}{2}$$

$$\Rightarrow \sin[\tan^{-1}(-\ln(x^2)] \to \sin\left(-\frac{\pi}{2}\right) = -1$$